AI Personal Finance Advisor Chatbot

1. Project Overview

AI Personal Finance Advisor Chatbot is a full-stack web application developed to help users manage and understand their personal finances in one place. The system allows users to record their income and expenses, create budgets, track savings goals, and receive AI-based financial guidance.

The main purpose of the project is to make personal finance management easier by combining traditional expense tracking with an AI-powered financial advisor. Instead of only displaying financial data, the application can analyze the available information and provide simple, understandable suggestions.

The application is developed using Python Flask for the backend, HTML, CSS and JavaScript for the frontend, SQLite with SQLAlchemy for database management, and Google Gemini API for AI-powered assistance.

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2. Problem Statement

Managing personal finances manually can be difficult. Users often lose track of their daily expenses, do not maintain proper budgets, and find it difficult to understand their spending patterns.

Traditional expense trackers mainly focus on recording transactions. They may not provide personalized explanations or suggestions based on the user's financial activity.

Therefore, this project aims to develop a simple web-based system that combines:

- Income and expense tracking
- Budget management
- Saving goal tracking
- Financial dashboard
- AI-powered financial assistance

3. Objectives

The major objectives of the project are:

- To develop a simple personal finance management platform.
- To allow users to record and manage income and expenses.
- To categorize expenses for better analysis.
- To provide budget management functionality.
- To track personal saving goals.
- To display financial information through a dashboard.
- To provide AI-generated financial insights.
- To create a professional and user-friendly web interface.

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4. Key Features

- **User Authentication**
  Users can register, log in, access their personal dashboard, and log out securely.
- **Dashboard**
  The dashboard provides an overview of important financial information such as total income, total expenses, balance, budgets, and savings.
- **Income Management**
  Users can add and manage income records by entering the source, amount, and date.
- **Expense Management**
  Users can record expenses with a title, category, amount, and date. Categories include Food, Rent, Transport, Education, Healthcare, Entertainment, Shopping, Bills, and Oth

- **Budget Management**
  Users can define spending limits for different categories and monitor their expenses against those limits.

- **Saving Goals**
  Users can create saving goals by entering a target amount and tracking the amount already saved.
- **AI Financial Advisor**
  The AI Advisor uses the available financial information to provide general financial suggestions and answer user questions. Google Gemini API is used for AI-based responses when available.
- **Rule-Based Fallback**
  If the AI service is unavailable because of API limitations or configuration issues, the application can still provide basic rule-based financial insights.

5. Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, Flask
Database	SQLite
ORM	SQLAlchemy
AI	Google Gemini API
Development	Visual Studio Code
Version Control	Git & GitHub
Deployment	Render

6. System Architecture

The application follows a simple client-server architecture:

User → Web Browser → Flask Backend → Database

For AI assistance:

User → Flask Backend → Gemini API → AI Response → User

The frontend provides the user interface, while Flask handles application logic, authentication, database operations, and communication with the AI service.

SQLite stores user-related financial information such as income, expenses, budgets, and saving goals.

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7. Database Structure

The application uses SQLite with SQLAlchemy.

The main entities are:

User

- User ID
- Name
- Login credentials

Income

- Source
- Amount
- Date

Expense

- Title
- Category
- Amount
- Date

Budget

- Category
- Budget amount

Saving Goal

- Goal name
- Target amount
- Saved amount

This structure allows the application to maintain organized financial records.

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8. Working of the System

The basic workflow of the application is:

1. The user creates an account.
2. The user logs into the application.
3. The dashboard displays the financial summary.
4. The user adds income and expenses.
5. The user sets budgets and saving goals.
6. The application calculates financial information from the stored records.
7. The user can interact with the AI Financial Advisor.
8. The AI provides general financial suggestions based on the available information.

9. Benefits

The system provides the following benefits:

- Centralized financial management.
- Easy income and expense tracking.
- Better understanding of spending categories.
- Budget monitoring.
- Saving goal tracking.
- AI-assisted financial insights.
- Simple and professional web interface.
- Can be accessed through a web browser.
- Can be extended with additional AI and analytics features.

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10. Future Scope

The project can be enhanced in the future by adding:

- Automatic bank transaction import.
- Advanced expense forecasting.
- Monthly financial reports.
- Budget and savings notifications.
- Mobile application.
- Advanced AI financial planning.
- PostgreSQL or another scalable production database.
- More detailed charts and financial analytics.

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11. Conclusion

The **AI Personal Finance Advisor Chatbot** combines personal finance management with artificial intelligence to provide users with a centralized platform for managing their financial activities.

The system allows users to track income and expenses, manage budgets, monitor saving goals, and receive AI-based financial guidance. The project demonstrates the practical use of **Flask, SQLite, SQLAlchemy, JavaScript, GitHub, cloud deployment, and Generative AI** in a real-world FinTech application.

The project provides a strong foundation for developing a more advanced intelligent personal finance management system in the future.

Project: AI Personal Finance Advisor Chatbot
Domain: AI
Backend: Python Flask
Database: SQLite
AI: Google Gemini API
Deployment: Render